

Project Overview

This project is an automated NIFTY options trading assistant built in Python. It gathers market data, option-chain data, volatility context, technical trend information, and derivatives positioning signals, then converts them into a simple trade decision: CALL, PUT, or NO TRADE.

The system is designed around a five-minute polling loop during market hours. It checks whether the NSE market is open, reads the latest NIFTY price and candle data, evaluates market conditions, selects an option strike, calculates trade levels, sends Telegram alerts, and records trades for later review.

The project is modular, with separate folders for data access, market analysis, AI-style decisioning, trade execution helpers, alert delivery, and configuration.

High-Level Workflow

1. Market gate: The bot first checks weekend status, NSE holidays, and market hours before doing any trading work.
2. Data collection: It fetches the NIFTY spot price, recent five-minute candles, India VIX, and the NIFTY option chain.
3. Market analysis: It identifies trend regime, support/resistance, breakout or breakdown conditions, volatility regime, gamma walls, liquidity sweeps, order-flow imbalance, and institutional positioning.
4. Confidence scoring: It combines trend, breakout, volatility, order flow, and positioning factors into a confidence score.
5. Signal decision: It generates CALL, PUT, or NO TRADE based on regime and breakout alignment.
6. Trade construction: It selects a nearby option strike, retrieves the option premium, calculates stoploss and target, sends a Telegram alert, and logs the trade.

Data Layer

Market data is sourced through Yahoo Finance for NIFTY spot price, five-minute NIFTY candles, and India VIX. The candle data is normalized into open, high, low, and close fields

- Regime detection uses moving-average relationship and ADX trend strength to classify the market as bullish, bearish, sideways, neutral, or unknown.
- Breakout detection compares the latest close with the previous twenty-period high or low to identify breakout, breakdown, or no-signal conditions.
- Open-interest analysis finds the highest call OI strike as resistance and highest put OI strike as support.
- Gamma wall detection identifies the call and put strikes with the largest open interest, giving a quick view of major option-chain pressure zones.
- Liquidity sweep detection flags cases where price takes out the previous candle high or low but closes back inside, suggesting a possible trap or rejection.
- Order-flow imbalance compares call-side and put-side traded volume to estimate directional pressure in the option chain.
- Institutional positioning compares aggregate put OI and call OI to classify positioning as bullish, bearish, or neutral.
- VIX filtering classifies India VIX into low, normal, or high volatility regimes so trade confidence can adapt to market environment.
- Performance analysis reads the trade log and summarizes total trades, signal distribution, breakout distribution, and regime distribution.

AI and Decision Layer

The signal engine intentionally keeps decisions simple and explainable. It avoids trades in

Strike selection rounds the NIFTY spot price to the nearest 50-point interval, then chooses one strike out-of-the-money for CALL or PUT trades.

Risk management calculates a fixed stoploss at 25 percent below the option entry premium and a target at 40 percent above the entry premium.

Trade logging writes each generated signal to a CSV file with time, direction, strike, entry, stoploss, target, regime, and breakout context. This creates a foundation for later performance tracking and strategy review.

Alerts and Automation

Telegram integration sends a structured signal alert containing direction, strike, entry, stoploss, target, regime, and breakout state.

The main bot loop runs every 300 seconds, matching the five-minute candle interval used by the analysis layer.

Configuration files define market hours, check interval, risk percentages, strike interval, NIFTY URLs, and news source URLs.

Technology Stack

- Core language: Python.

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- The main workflow currently calculates confidence before breakout, order flow, and institutional positioning are assigned. Those analyses should run before the confidence score is calculated.
 - The mock option-chain fallback includes option premiums but does not include all open-interest and traded-volume fields needed by some analysis modules. The mock data should be expanded for complete offline testing.
 - The signal engine does not yet use sentiment, liquidity sweep, gamma walls, support/resistance, or VIX directly in the final CALL/PUT decision, although those signals are already available elsewhere in the project.
 - Telegram credentials are loaded from a configuration module. In a production version, these should be handled through environment variables or a local secrets file excluded from version control.
 - Exception handling exists in several data modules, but the full workflow would benefit from stronger validation when live data is incomplete or delayed.

Suggested Next Enhancements

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- Reorder the main analysis pipeline so every confidence input is calculated before

Project Status Summary

You have built a working modular foundation for a NIFTY options signal bot. The project

already covers market access, option-chain interpretation, technical regime analysis,

volatility filtering, flow/positioning analytics, trade construction, Telegram notification,